

CHITRANG POTDAR

Backend Engineer | Applied AI

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PROFESSIONAL SUMMARY

Backend Engineer with a focus on Applied AI. Experienced designing REST APIs, autonomous workflows, multi-tenant architectures, and AI-enabled services with focus on maintainability and reliability. Built developer tooling, RAG pipelines, automation systems, and cloud-native applications using modern Python technologies. Delivered systems spanning investment platforms, AI pipelines, LMS/CMS, and full-stack productivity software.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL

Backend: FastAPI, Flask, REST APIs, Celery

Databases & Caching: PostgreSQL, Redis, MSSQL

AI & Agent Frameworks: Claude API, LlamaIndex, LangChain, LangGraph, MCP, AWS Transcribe, Google Cloud STT, Whisper

Cloud & DevOps: AWS (Lambda, API Gateway, S3, EC2), Docker, GitHub Actions, CI/CD

Third-Party Integrations: SendGrid, Vimeo, Web Push (VAPID)

Observability: Grafana, Loki, Promtail, LangSmith

Testing & Automation: Pytest, Playwright

ML & Data: Scikit-learn, Pandas, NumPy, Polars, Seaborn, Matplotlib

PROFESSIONAL EXPERIENCE

Backend Engineer · ArrayPointer Pvt. Ltd., Pune (Oct 2025 – Present)

Investment Operations Platform

Flask · PostgreSQL · Redis · Docker · GitHub Actions · AWS EC2 · Grafana · Loki · Promtail · Pytest · Playwright

- Built a multi-tenant Flask backend with query-level data isolation across users, portfolios, and organizations on a shared PostgreSQL schema; added Redis session caching to reduce redundant DB round-trips, maintaining 80%+ Pytest coverage.
- Developed REST APIs for portfolio review, rejected order tracking, and financial calculators; integrated Jira API to automatically create issues from application events, removing manual reporting.
- Automated NSE rejected order processing via Python API integrations and Playwright-driven end-to-end client workflows, saving ~2–3 hrs/week of manual effort.
- Set up the observability stack: Promtail shipping logs to Loki, with Grafana dashboards providing structured visibility into service health and error trends.
- Built a domain-specific MCP server exposing investment platform data to LLM clients, annotating tools/functions with domain terminology so models resolve business concepts without raw schema knowledge.
- Implemented async email delivery via Redis + Celery and VAPID-based Web Push for real-time single-user alerts, including UI failure notifications for unsuccessful email delivery.
- Extended task management with hierarchical ownership/access control across organizational levels; built modules for task reporting, employee performance analytics, and CRM workflows.
- Added authenticated user session tracking and a calendar view covering holidays, meetings, and out-of-office schedules, with task/CRM API restrictions enforcing workflow and availability rules.

LMS & CMS Platform — Jaro Education

FastAPI · PostgreSQL · SendGrid · Vimeo · AWS S3 · Modular Monolithic Architecture

- Designed a modular-monolithic LMS/CMS backend using FastAPI and PostgreSQL, supporting 10K+ students at ~300 RPS, with SendGrid email-verified authentication.
- Built admin-side course creation and student management modules, integrating Vimeo for video upload/streaming.
- Developed student-facing APIs for course access and certificate retrieval, with AWS S3 backing certificate storage.

Conference Management CMS — IISc Bangalore

FastAPI · PostgreSQL

- Built a conference management system supporting conference publishing, participant registration, admin approval/rejection workflows, and attendance tracking.

Stocks Analysis & Decision System

Flask · PostgreSQL · React · Pandas · Docker · Global Data Feed API

- Integrated GDF APIs for EOD and 15-minute delayed intraday data; built Flask REST APIs and a React frontend for market data and stock analysis.
- Persisted historical data in PostgreSQL and added scheduled EOD processing to avoid repeatedly fetching and processing historical data.
- Automated EOD ingestion and indicator calculations through background workflows, moving expensive calculations out of request paths to improve response times.

MahaNRI — NRI Social & Community Platform (Backend Lead)

FastAPI · PostgreSQL · React · AWS · Docker

- Developed backend services for a Maharashtra Government-project social platform targeting the NRI community across multiple applications sharing a common backend.

- Designed and implemented REST APIs across application modules while coordinating API contracts and integration requirements with frontend teams.
- Led backend development activities across a team of developers/interns, including task breakdown, technical guidance, code reviews, PR management, and merge decisions.
- Participated in client discussions to clarify requirements, evaluate technical approaches, and translate product requirements into backend implementation tasks.
- Managed backend deployments and database migrations while coordinating development changes across multiple application modules.

PROJECTS

SOLID-Agent-Harness — Multi-Agent Code Analysis Tool

Node.js · LangChain · LangGraph | github.com/Dev-Chitrang/SOLID-Agent-Harness

- Built a language-agnostic CLI developer-tooling platform running deterministic, graph-based LLM workflows for repository analysis, documentation generation, and code quality — six individual commands (audit, bugs, docs, readme, review, explain) plus five composite workflows.
- Implemented a critic-loop pattern per graph (Agent → Critic → conditional retry) that exits on a quality score threshold or iteration cap, improving output reliability over a single-pass agent call.
- Designed composite commands using LangGraph's parallel fan-out with a barrier-join summary node, running multiple analysis graphs concurrently and synthesizing results into one executive report.
- Built a provider-abstraction layer supporting OpenAI, Anthropic, Gemini, and NVIDIA, swappable without changing any graph code, plus an interactive TUI for provider/command/model selection.
- Architected registry-driven command dispatch so new commands can be added via a folder and registry entry alone, with no changes to core harness or TUI files.

Unified Workspace — Full-Stack Productivity Suite

FastAPI · React · PostgreSQL · Redis · Celery · Docker | github.com/Dev-Chitrang/Productivity-SaaS-Ful-Stack-App

- Built a self-hosted productivity platform unifying task management, notes, calendar scheduling, WebRTC audio meetings, collaborative whiteboarding, and AI meeting analysis into one application.
- Designed a layered Repository → Service → Controller architecture with dependency injection throughout the FastAPI backend for testability and separation of concerns.
- Built a Redis-backed meeting engine with session-scoped lifecycle management and WebSocket signaling for real-time audio meetings.
- Integrated AI-powered meeting analysis via NVIDIA NIM (Llama 3.3 70B) to generate summaries, agenda coverage, and actionable task suggestions from meeting transcripts.
- Implemented JWT authentication with Redis-backed token revocation, OTP-based two-factor auth, and Google OAuth sign-in.
- Maintained 100+ test files with 2,100+ test cases using pytest-asyncio; deployed as a 6-service Dockerized stack.

EDUCATION

B.E. Computer Engineering — 8.65 CGPA
Savitribai Phule Pune University, Pune | 2021 – 2025

CERTIFICATIONS

- IBM AI & Cloud — AICTE Certified
- VOIS AI & Data Analysis — AICTE Certified

RESEARCH & PUBLICATIONS

Corporate Credit Rating Analysis using AI & Sentiment Analysis
IIRCCCE | Impact Factor: 8.65 | Nov 2024 | github.com/Dev-Chitrang/Final-year-project/tree/Backend